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Modular Design for Adaptive Sports Equipment: A Rowing Ergometer Case Study

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Abstract: This paper presents a modular design workflow for personalised sports equipment balancing athlete-specific customisation and system-oriented standardisation for existing equipment platforms. A case study of a customised rowing seat for a paralympic athlete forms the foundation of the approach. Integrating athlete 3D-scanning, rapid prototyping, production through 3D-printing and composite reinforcement allow designers to benefit from insights made throughout the design process. The proposed approach offers a replicable framework for inclusive design in adaptive sports and rehabilitation.

Keywords: Adaptive sports technology, custom equipment, additive manufacturing, inclusive sports design, paralympic equipment

1 Introduction

Consider the challenge of training with footwear that is oversized and misaligned with the anatomical contours of the foot. While tightening laces, adding extra padding, or tape may provide temporary improvement, these modifications compromise performance and optimal training results will remain unattainable. This analogy reflects the experience of many athletes in adaptive sports, whose performance potential is constrained by equipment not designed for their specific needs (Jamieson & Wijesundara, 2025). Conversely, equipment for paralympic athletes is often custom designed to accommodate individual needs. This customisation process relies on expert knowledge, where specialists draw prior experience to identify specific design solutions for each athlete. While such customisation tailoring ensures optimal functionality, it also leads to high production costs. The development of paralympic athletes is constrained by financial, material, and skill-related requirements (Khurana et al., 2021). Challenges are also evident in sport facilities, where the infrastructure can become an obstacle for paralympic athletes (Sobiecka et al., 2019). Standard gym equipment often fail to meet the needs of athletes with disabilities, making the equipment unusable (Calder et al., 2018). Athletes may need to rely on facilities suitable for adaptive sports, which are often less available. This limitation introduces an environmental barrier to participation, namely increased transportation difficulties (Jaarsma et al., 2014; Diaz et al., 2019). Additionally, dependence on others during the training process has been identified as a barrier that can hinder both training and athlete performance. As Macías et al. (2022) highlights, the importance of psychological factors such as self-esteem, motivation, and stress reduction, together with physical and technical components, play a crucial role in shaping the effectiveness of the training process.

This study will focus on addressing these issues through a case study on adaptive rowing. The physical demands and potential for adaption shows opportunities for bridging the gap between customised equipment and compatibility with standard gym equipment. Technical and conceptual challenges have been identified when training with rowing ergometers, both related to the learning curve but also adjustments to personal use where assistance is required to set it up. Difficulties with adjustment mechanisms have also been a limitation for an effective training process (Anderson et al., 2024). Wong et al. (2022) identified equipment-related issues as the most frequently reported barrier to the use of adaptive rowing ergometers. Specifically, inadequate equipment accessibility and dependence on staff assistance were identified as common concerns. This dependence reduces accessibility and restricts the feasibility of training in standard gym facilities. Improving the portability and ease of installation of customised equipment would make training in standard gyms more efficient and accessible for adaptive athletes. Previous research has shown that the performance of adaptive rowers during ergometer rowing training can be enhanced by adjusting seat and backrest position (Eikevåg et al., 2020; Severin et al., 2021). Furthermore, the development of form-fitted and lightweight paralympic equipment through 3D-scanning of athlete geometry and 3D printing components demonstrates the potential for rapid production alongside improved ergonomic fit and stability (Nesheim et al., 2024).

The aim of this article is to explore a modular, human-centred design approach that integrates two seemingly opposing design philosophies. The approach balances user-specific customisation, enabling high-performance training with a generalised framework that ensures compatibility with standard gym equipment. This work contributes to the development of inclusive and adaptable training environments that bridge the gap between customised and standardised equipment design.

2 Design Context and Approach

The workflow adopted in this study is inspired by the Research through Design (RtD) approach, where knowledge is generated through the process of designing and reflecting on processes, materials, and artefacts (Godin & Zahedi, 2014). Rather than testing a predefined hypothesis, we aim to present the exploration of how design methods can function as a bridge between two seemingly conflicting concepts, customisation and standardisation. The system-driven approach for standardisation is often viewed as efficient and generally easier to adopt than customisation (Scott-Sharoni et al., 2021). It is often characterised by systems that remain adaptable, repeatable, and broadly applicable in their respective context. Customisation on the other hand, focuses on designing for individual ergonomics and experiential needs, but simultaneously carries the connection to long design processes and expensive end products.

This study aims to shed light on the fact that a product design can benefit from combining qualities from both design philosophies discussed above. We exemplify the merging of the concepts through a mixed approach to sport equipment design, specifically in rowing ergonomics for paralympic athletes. The approach aims to maintain the highly important athlete customisation aspects of the product design, while simultaneously generalising for standardised sports equipment widely used in the market today. The process emphasises reflection-in-action (Schön, 1986), where each process step provides insights that are taken into considerations in the further design choices, progressing through digital representation, physical fabrication, and user interaction.

The development of adaptive sports equipment often takes place in resource intensive, highly individualised settings. Athletes rely on uniquely constructed setups manufactured by specialised personnel. This equipment is often custom end-to-end, thus requiring transport of bulky objects and dedicated space in each location the athlete travels to. When removed from their main training hubs, this dependence can make training difficult or even impossible. This limitation highlights a broader challenge in adaptive sports equipment design: The lack of modular solutions that enable athletes to maintain training routines across various environments. Addressing this gap requires a design approach which is open to combining aspects from both system-oriented platforms and individual customisation.

The physical context for the design is a modular adaptive rowing seat, developed to easily mount onto a widely used rowing ergometer, the Concept2 RowErg (Concept2, 2025). Previous work on the project includes a study on data-driven design for customised rowing ergometer seat positions (Eikevåg et al., 2020). Further work has included vision-based dynamic interaction studies on rowing ergometer equipment (Johnston et al., 2022). In this study, we shift the focus from performance measurement to design process insights, investigating how designers can structure workflows that:

1. Integrate body-specific digital data into customised components.
2. Merge custom features with standardised interfaces for employing existing platforms.
3. Enable rapid iterations and in-action insights throughout the design process.

3 Case Study: Customised Rowing Seat on Standardised Platform

To investigate the possibilities for a unified customisation and standardisation design pipeline for paralympic sports equipment development, a case study was conducted on a common rowing ergometer, available in most standard gyms today. The aim of the case study was to identify interfaces between off-the-shelf sports equipment and from there define the need for bespoke solutions for the paralympic athlete. The study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Norway (ID: 661300, approved 16 January 2025). Informed consent was obtained from all subjects involved in the study.

3.1 Customised Rowing Seat Development

Figure 1 shows the development process of the customised rowing ergometer seat. All steps of the iterative design and production after the initial 3D-scan were carried out by the authors. The initial steps of the process include 3D scanning of the athlete body and a standard seat rest, followed by the combination of these elements in digital surface modelling software. The specific angles chosen for the seat rest and incline of the lower seat are derived from results of user tests conducted using the low-cost parametric rowing seat presented in (Eikevåg et al., 2020). Before moving into full scale production of the custom mesh structure, a series of small-scale 3D-printed iterations aided the designer and athlete with a tactile feel for the scanned surfaces.

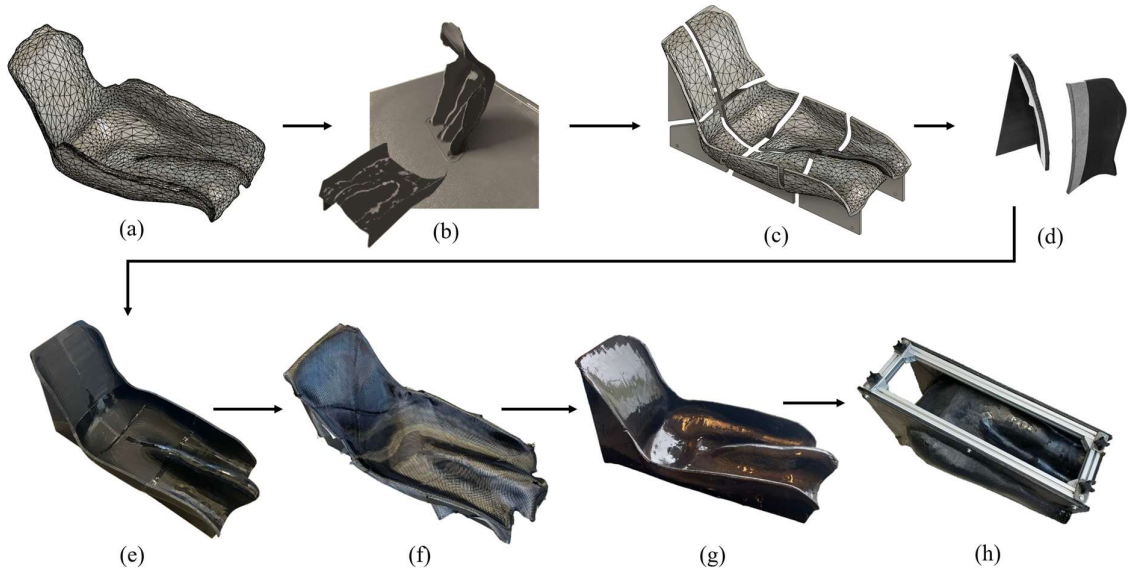


Figure 1: Design process of the custom rowing ergometer seat: (a) 3D-scan mesh preprocessing, (b) small-scale 3D-printed prototypes, (c) object splitting into non-supported printable parts, (d) 3D-printed subparts, (e) glued assembly of printed shape structure parts, (f) carbon fibre layering, (g) epoxy laminating with carbon fibre to enclose the printed shape structure, and (h) assembly on to the system-compatible frame interface

The next steps of the design process involved making the interfacing between the seat mesh and the standardised mounting frame, and splitting the resulting structure into parts that are printable on a commercially available consumer grade 3D-printer (Creality, 2025). After printing the parts in Polylactic Acid (PLA), they were glued together using epoxy. The assembled structure serves mainly as the shape and form defining component, whereas the following carbon-epoxy laminating step provides strength and robustness. Material choices were made with availability and cost in mind. Maintaining a low weight ensured that the paralympic athlete could independently move and handle the bespoke equipment. Finally, the customised seat structure was mounted onto the system-adapted aluminium frame. The mounting frame of the rowing seat was constructed from standard t-slot 30x30 aluminium profiles. The quick-release mounting clamps were made from off-the-shelf 6mm aluminium plates, and thumb screw heads were 3D printed in the same PLA material as the inner structure of the personalised seat. All the above-mentioned design choices were made to optimise for reproducibility, thus lowering the thresholds for adopting the same mounting concept for other athletes. Figure 2 shows a selection of the prototypes (P1-P5) for the system-compatible mounting interface with the rowing ergometer. P3-P5 are iterations of the final clamping system, with gradual improvements of ergonomics for the thumb screws and material changes from Medium Density Fibre (MDF) plates to 6mm aluminium plates. To avoid increase in cost, no mechanical processing such as Computer Numerical Control (CNC) machining was utilised.

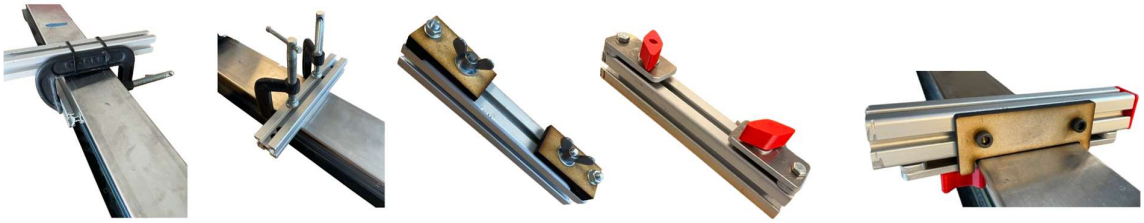


Figure 2: Quick mounting interface prototyping iterations: P1 and P2 basic RtD prototypes for exploring the solution space, P3 and P4 iterations of the sliding clamp, P5 final sliding clamp with a guide rail for centring the seat on the rowing ergometer

The final mounting frame and quick release clamp mounting can be seen in Figure 3 (a). A side view of the quick-release clamp mounting system is shown in Figure 3 (b) and (c), where the latter shows the clamp after it has been slid onto the concept2 rowing ergometer beam and tightened with the thumb screw.

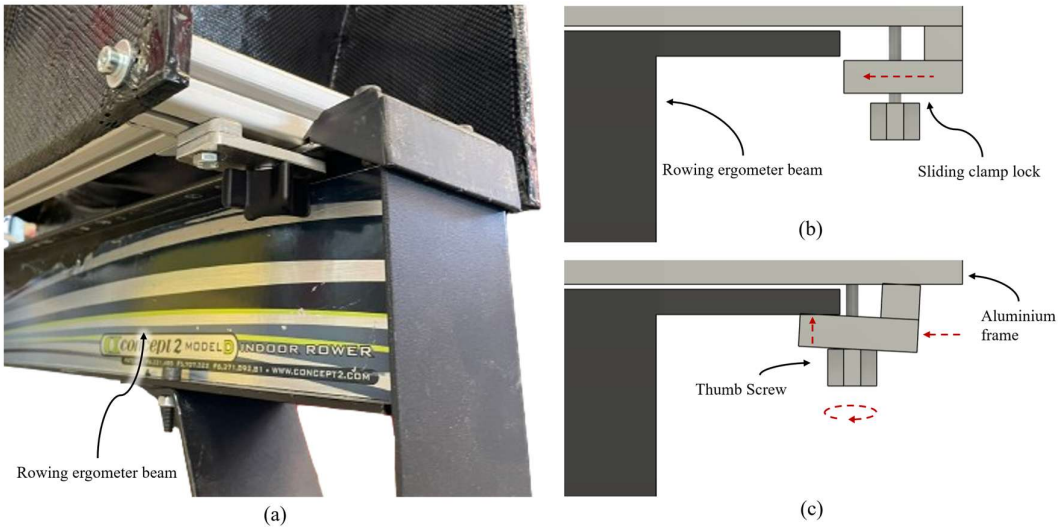


Figure 3: Standardised quick release mounting for the Concept2 rowing ergometer: (a) seat mounted on the rowing ergometer, (b) seat clamp lock in open position, and (c) seat clamp lock in closed position

3.2 Footrest Platform Development

During the design and development process of the customised rowing seat, a need for adaptation to the footrest became apparent. This component was designed on request from the athlete, due to their specific need for a surface to mount a set of shoes onto, allowing them to keep their feet in place given a paralysed lower body. Additionally, the system-adapted mounting frame of the seat adds height to the seating position, which should be matched for the footrest. The development process for the footrest was similar to the seat but simplified as much as possible to maintain low cost. The material choice was made to not be restricted to mounting rails or slots. The shoes were mounted directly to the wood plate using screws and provided sufficient support for the athlete. The mounting choice was based on data collected on athlete preference for foot support. No further reinforcement was needed, as minimal forces were acting through the footrest other than the weight of the legs of the athlete. Figure 4 shows the process from initial MDF prototype through digital design into final product.

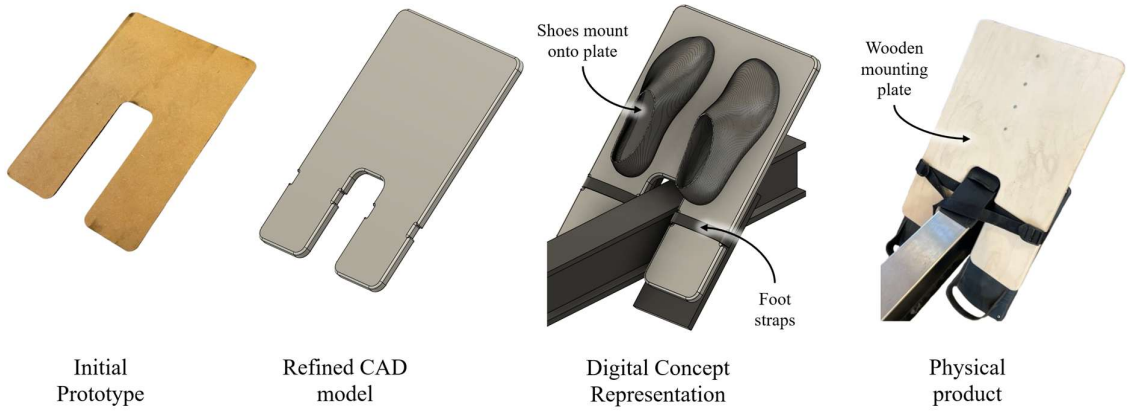


Figure 4: Footrest Platform from initial prototype to digital model and final product, mounted onto the rowing ergometer using the existing foot straps

3.3 Full System User Test

Figure 5 displays the seat and a custom footrest mounted onto a standard Concept2 rowing ergometer without the need to do any modification to the standard equipment. The new seat provides the necessary support and freedom of movement to allow the paralympic athlete to perform and stay comfortable during their training. Note that Figure 5 is provided for illustrative purposes, with the athlete replaced to protect the privacy of the primary user. After design and production, the seat has been taken into regular use and proven durable in testing with several athletes in addition to the main user. The athletes report the weight and size of the seat to be of great importance to the handling and convenience in their daily

routines. The athlete emphasises that not needing to alter the standard rowing ergometer to be able to fit their bespoke equipment is a key factor to the ease of use. Testing of the mounting process shows that the athlete is able to transport and efficiently mount the equipment onto the existing standard platform without the need for assistance.



Figure 5: User testing of finalised product quick mounted on to the rowing ergometer

3.4 Design Case Study Insights

The case study has provided several valuable insights for consideration into the formulation of the balanced approach proposed in section 4, both from user interviews and reflections from the designers. They are condensed into a brief set of points listed below:

1. Processing of the mesh data benefitted from the use of small scale early-stage 3D-printed prototypes, which provided a tactile experience of the surfaces.
2. High resolution 3D scans of the athlete provided a false sense of precision in the early stages of mesh processing. This was discovered through the utilisation of small-scale prototypes.
3. A stable and simple standard frame interface allowed rapid iterations over the customised components. The modular design allows each new user to become a reconfiguration, not a complete redesign.
4. The training equipment platform served as one of the main design constraints. Investing time in a solid and efficient interface between personalised and standardised subsystems is beneficial.
5. Consumer-grade 3D-printing and modest material choices allowed a lower threshold for iterations on the personalised subsystem and can support wider adoption of similar equipment for other athletes.
6. Throughout the design process, the need for balancing aspects of customisation and standardisation became increasingly apparent.
7. The athlete describes the new seat as a game changer in terms of making training accessible for them, especially in a travel context.

4 Customisation - Standardisation Design Approach

Based on the experiences and insights gained through the case study on a customised-standardised platform for a common rowing ergometer, we propose an approach which aims to draw benefits from both the system-oriented mass production design philosophy and bespoke inclusive design. The main aim of the proposed approach is to customise where it is absolutely necessary, while generalising and maintaining system compatibility when possible. By doing this, we aim to lower cost compared to conventional orthotics development, which is often fully custom made, reduce production time, and allow reflection-in-action to aid the design process through incremental small-scale prototypes. The proposed pipeline is outlined in Figure 6 below.

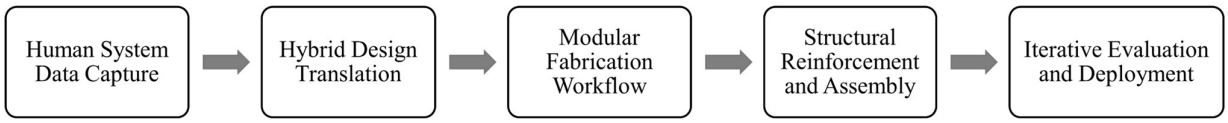


Figure 6: Design approach for balancing customisation and standardisation for adaptive sports equipment

The workflow is initiated with gathering of necessary information and usage data about the athlete and the target equipment platform. By 3D-scanning the body of the athlete and collecting information on their subjective equipment preferences, we establish the requirements for personalised ergonomics. In parallel to the personalised data-capture, we also need to gather information on the target equipment platform to define the design constraints for standard interfacing. Capturing these datasets in parallel is done to motivate the designer to equally consider the dual design interests in later design decisions.

Hybrid design translation involves the process of merging the requirements and features from the two datasets in step 1 to make a coherent product. In addition to processing the mesh data from athlete scans and designing an efficient system mounting interface, the negotiating process between the standardisation design constraints and the personalisation is of great importance. Small scale physical prototypes can aid the designer and athlete in their collaboration by providing physical object intuition and reflection-in-action processes for the designer.

To be able to produce the resulting design without the need for large specialised and costly equipment, a modular fabrication workflow is adopted in the third pipeline step. Subcomponents are sized to be printable on consumer-grade 3D printers in commonly used materials to minimise costs. Standard hardware such as aluminium profiles and fastening hardware are specified and dimensioned to the required safety factors.

Structural reinforcement and assembly bring the components together to a functional product. Printed elements are glued together and function as the shape defining structure for the athlete interfacing surfaces. A composite fibre epoxy reinforcement is applied to provide functional strength and surface finish to the product while maintaining low overall weight to ease handling of the product. Finally, the customised subsystem of the assembly is mounted onto the standardised mounting frame which adds further structural integrity in addition to quick mounting capabilities.

Iterative evaluation and deployment is the final step of the approach. This step involves an evaluation loop where the coaches and athletes generate feedback and insights together with the designer. Due to the nature of the dual design philosophy, new iterations and tweaking of the personalised part of the product benefit from the fact that the main platform and interface remains constant. This saves time and production effort and allows easier adaptation to new athletes who can benefit from the standardised part of the design, as well as the insights gained in previous personalisation design processes.

5 Discussion

The workflow presented in this study exemplifies a human-centred design approach leveraging a combination of customisation and standardisation to obtain a hybrid workflow. Rather than choosing between bespoke one-off solutions and mass-produced products, the approach uses a personalisation design approach where it is needed, and system-oriented solutions where the design constraints allow it. Combining these aspects allow us to achieve a duality between performance and low overall costs for the athlete.

Traditional engineering design is often characterised by the adoption of one specific design philosophy. Through the case of sports equipment design for paralympic athletes, we demonstrate that a combination of customisation and standardisation can lead to a more efficient design process. Personalised components are built to fit to a stable and standardised frame, enabling rapid reconfiguration and adaptation to new athletes without requiring a complete redesign. This approach can be particularly valuable in domains like adaptive sports and inclusive design, fields characterised by

high user diversity and needs for system integration. The addition of platform compatibility provides an edge compared to traditional fully custom equipment due to the fact that the athletes can utilise existing infrastructure and equipment available to able-bodied athletes. The use of consumer-grade tools and materials makes the concepts available to a wider community of designers and users.

The design approach aims to be a replicable framework for future applications, both in paralympic sports equipment development and other relevant fields. In each step, an emphasis is placed on the balancing of the two main interests of customisation and standardisation. An important tool in the early stages of the development process has been the use of small-scale prototypes and physical representations of the design. Although important to the outcome of the design process, these low effort iterations have not been assigned to a designated step in the design pipeline. Rather, we encourage the constant use of such prototypes in all stages of the development process. Such prototypes may aid in early detection of design weaknesses and limitations. The cases of the seat and footrest demonstrate some important aspects of the proposed approach, through their differences in complexity and size. The rowing seat includes a high degree of athlete-specific customisation, whereas the footrest customisation is to a larger extent based on the data collected on athlete preference. Maintaining simplicity where possible is as mentioned an important priority in the proposed pipeline. By using the foot straps already present on the mass-produced piece of equipment, we reduce complexity and weight of the custom equipment. Considerations of predicted loads and base material choice allowed us to avoid additional reinforcement of the footplate. The steps of the proposed pipeline should always be considered in light of the size and complexity of the envisioned product.

Although the modular design approach shows considerable promise for personalised equipment development, certain limitations warrant attention. In the initial steps of the pipeline, data collection is performed by 3D-scanning the athlete. During this process, a high resolution might lead to a false sense of precision. Mitigating this tendency can be done through actively investigating the mesh for irregularities in the processing step but also through utilising small scale printed prototypes which provide a tactile experience of the surfaces. Furthermore, the choice of materials reflects the desire to maintain a low cost and lightweight solution. For structures that require a higher degree of structural strength and/or durability, more sophisticated materials might be preferred. For larger production volumes, material separability for recycling after end of product lifetime should also be considered. The design approach aims to address the threshold and cost of bespoke equipment by applying production methods and tools more widely available in the engineering community. Although we believe that the threshold for paralympic equipment development is lowered in our case, it is important to note that a certain set of skills is still required, and tacit experience is beneficial for the designer. These reflections underscore the importance of iterative design validation and learning throughout the design process.

6 Conclusion

In this study, we have presented a modular, human-centred design approach for personalised sports equipment adapted to commonly available sports equipment platforms. The pipeline is derived through the design and production of a rowing seat adapted to a paralympic athlete and quick-mounted to a widely used rowing ergometer. By combining personal ergonomic data with system-oriented design constraints, we have demonstrated a way to customise where it is absolutely needed, while maintaining standardised features where possible. We capture and illustrate a replicable framework using consumer-grade tools and materials, allowing access for a broader community. Applying the presented design principles can aid more adaptive athletes in their needs and substantially increase the number of hours spent in conventional gyms. Overall, we believe the presented design approach opens new possibilities in inclusive design, rehabilitation, and adaptive sports equipment, where diversity in user needs meets a demand for system compatibility.

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